

# Heaven Alakraa

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## EDUCATION

**Brown University, Sc.B in Mechanical Engineering, GPA: 3.96**

**Providence, RI | Aug 2024 - May 2028 (Expected)**

**Relevant Courses:** Engineering Statics and Dynamics; Mechanics of Solids and Structures; Computer Aided Visualization and Design; Biomaterials; Thermodynamics; Fluid Mechanics; Electrical Circuits and Signals; Control Systems Engineering; Electricity and Magnetism; Biomechanics.

**Awards:** American Society of Biomechanics, 2026 Carlo De Luca Biomechanics Summer Undergraduate Research Experience (B-SURE) Award.

## TECHNICAL SKILLS

- CAD & Design: SolidWorks, Fusion 360, Onshape
- Engineering Drawings, Simulations, Manufacturing
- 3D Printing, Rapid Prototyping, Woodworking
- Coding & Data Analysis: MATLAB, Python
- Arduino, LTsprice, ImageJ, COMSOL
- qPCR, ELISA, DNA methylation analysis

## PROFESSIONAL EXPERIENCE

**Bioengineering Laboratory, Orthopedics Dep. at RIH, Engineering Research Intern**

**Providence, RI | May 2026 – Present**

- Design active and passive pendulum testing apparatus in SolidWorks to measure coefficients of friction in biological joints.
- Fabricate and assemble an experimental testing device in the RIH Orthopaedic Foundation testing facility; conducting mechanical validation to characterize tribological properties of soft tissue structures under physiological loading conditions.

**Lilac Biosciences Inc., Product Design Intern**

**Providence, RI | Jan 2025 – Present**

- Redesign RNA purification tube geometry in SolidWorks to replace centrifuge-dependent separation with integrated filtration; fabricating and iterating prototypes via 3D printing for experimental validation.
- Evaluate filtration membrane candidates through structured literature review and vendor analysis to identify optimal pore size and material compatibility for RNA retention.

**Tripathi Lab: Biomedical Engineering at Brown University, Research Assistant**

**Providence, RI | Jan 2025 – Sep 2025**

- Quantified AAV capsid and transgene concentrations via NanoMosaic's Tessie nanoimaging instrumentation.
- Independently developed and tested plate reusability protocols using chemical and enzymatic stripping agents to enable nanoplate reuse, reducing per-experiment consumable costs.
- Supported robotic liquid handling validation by analyzing automated assay performance against manual benchmarks.
- Performed DNA extraction, bisulfite conversion, BioAnalyzer QC and DNA methylation analysis for epigenetics research.

**Brown University School of Engineering, Teaching Fellow**

**Providence, RI | Aug 2025 – Jan 2026**

- Supported instruction for 200+ students in introductory engineering design through the National Education Opportunity Network; led weekly design brainstorming and prototype testing sessions, held weekly office hours, and delivered feedback.

## PROJECTS

**Can Opener Mechanical Design & Analysis (SolidWorks)**

- Designed a fully constrained multi-component can opener assembly in SolidWorks, producing detailed 2D engineering drawings, mated assembly, exploded views, and a complete drawing package for all components.
- Conducted static FEA on components under operational loading, validating my design against stainless steel's yield strength.
- Performed motion analysis on the gears, quantifying mechanical advantage and contact force distribution across a full cycle.

**Cantilever Beam Deflection Scale (MATLAB, Brown Design Workshop)**

- Designed and fabricated a cantilever beam-based mechanical scale, applying Euler-Bernoulli beam theory and Hooke's Law to translate analytical deflection expressions into a physical model using band saw, drill, and other wood tools in the BDW.
- Optimized beam dimensions and load placement using MATLAB, computing spring constant, bending stress, and safety factor; experimentally validated design through incremental calibration, predicting unknown masses with <10% error.

**Next-Gen Material Selection Vertebral Body Replacement Case Report (Ansys Granta Edupack and ASM Medical Materials)**

- Translated clinical needs and finite element analysis (FEA) into quantitative mechanical and biological design criteria.
- Identified Polyetherimide (PEI) as a leading candidate for future clinical application by screening 4,395 materials in database.

**Heart Rate Sensor & Formula 1 Safety Gear Testing**

- Prototyped a real-time heart rate monitor using MAX30102 pulse oximetry and Arduino OLED display; analyzed and visualized BPM data in MATLAB across simulated Formula 1 safety gear conditions, presented at an engineering showcase.

## LEADERSHIP & COMMUNITY DEVELOPMENT

**Brown Formula Racing Team (Brown FSAE), Cooling and Radiator Testing Engineer**

**Providence, RI | Sep 2025 – Present**

- Designed wind tunnel test rig in SolidWorks for radiator characterization; sourced piping and fluid system components through McMaster-Carr to support fabrication of the cooling system test apparatus.
- Evaluated radiator thermal performance via 3-bucket calorimetry and mass flow rate testing, collecting experimental data to validate heat transfer coefficients and inform cooling system redesign.

**Student Accessibility Services, Operations and Administrative Coordinator**

**Providence, RI | Mar 2025 – Present**

- Lead front-line support for 7000+ students, improving process efficiency and service delivery under tight academic deadlines.